

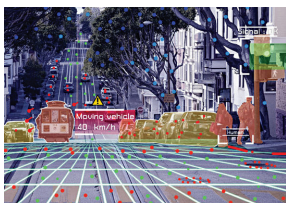
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Monocular 3D thermal ranger

The 3D Thermal Ranger from Owl AI is the world's first monocular 3D thermal ranging solution. It offers up to 150x improvement in resolution and cloud density compared to most lidar solutions available in the market. The



module can improve the safety of autonomous vehicles because of its range,

resolution, and ability to detect the speed and direction of an object even in extreme weather and during low lighting conditions. Unlike most other solutions, the 3D Thermal Ranger requires a single module to detect objects throughout the entire field of view and is immune to vehicle vibration for reliable and robust mapping, thus saving space and energy. The module uses AI deep learning and foveal optics to extract dense range maps from radiated heat signatures without illumination. Moreover, it can detect and identify objects, which can enhance path planning for self-driving vehicles.

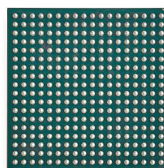
Owl AI

<https://www.owlai.us>

Low-power FPGA module

GateMate A1 FPGA is one of the smallest FPGA components featuring 20,480 logic elements, making it suitable for low-powered applications. The GateMate A1 FPGA requires only two supply voltages, making it easier to run and integrate into a product. The FPGA is suitable for use in both small-scale projects as well as high-volume applications as it offers high logic capacity,

multi-die capability, and a low power footprint at a competitive price, making it ideal for low-cost projects. The architecture of GateMate is based on the Cologne Programmable Element, allowing efficient construction of various logic configurations, including multipliers for digital signal processing. The FPGA offers over 162 GPIOs for fast communication and features configurable single-ended or LVDS differential pairs, supporting double data rate (DDR). It also features embedded



Serializer-Deserializer (SerDes) blocks that enable high-speed serial communication at up to 5Gb/s. The GateMate FPGA devices incorporate open source design tools and a proprietary place and route tool by Cologne Chip, providing easy access to innovation potential. Additionally, GateMate FPGAs benefit from reliable and advanced process nodes (Globalfoundries 28nm SLP).

Cologne Chip

<https://www.colognechip.com>

DFB laser source

Moonstone by Lightelligence is a high-power, multichannel, single- or multi-wavelength DFB laser source. It simplifies optical packaging assembly through an automated process and comes in a compact package, saving space. It enables the designers to replace 4 laser diodes with a single Moonstone module without affecting the accuracy. Moonstone incorporates off-the-shelf laser chips, utilising techniques like eutectic soldering, die-bonding, and wire-bonding.

The device offers single- and multi-wavelength options, employing either free-space coupling or a beam combiner for fibre connection. Precise thermal control is achieved by the module with a thermoelectric cooler and thermistor.



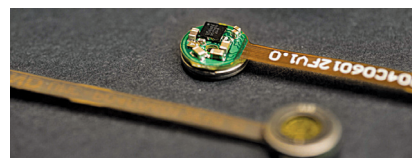
Moonstone features compact-size high-power DFB laser chips, low-cost production, and low process requirements. Its customised packaging ensures reliability and affordability while supporting up to 8 wavelengths, which enables flexibility in multi-wavelength applications. With high output power and low laser-to-fibre coupling loss, Moonstone excels in data communications and lidar systems. Its compact design, precise thermal control, and versatile packaging make it efficient and reliable. Moonstone can be used in various applications, such as lidars, test and measurement equipment, and data communication.

Lightelligence.Ai

<https://www.lightelligence.ai>

Thin MEMS speakers

The USound's Kore 4.0 UMC-P04C06012F is the thinnest MEMS speaker on the market. The module is a ready-to-use solution that can be integrated with any Bluetooth system-



on-a-chip (SoC). The Kore 4.0 speakers consume around 0.9mA current when producing 75dB sound, making them extremely energy efficient. The small size and lower energy consumption of the Kore 4.0 will enable designers to